Tanvir Ahmed Khan

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Research Portfolio • Google Scholar • GitHub • ORCID

Education

North South University Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering 2021 – 2025

- **CGPA: 3.80/4.00** | *Summa Cum Laude*
- Relevant Coursework: Machine Learning, Pattern Recognition, Numerical Methods, Probability and Statistics, Introduction to Linear Algebra, Design and Analysis of Algorithms

Research Interests

Multimodal learning; retrieval, memory, and evidence-aware reasoning; video understanding; 3D vision and neural rendering; audio-driven facial animation; low-resource natural language processing.

Ongoing Research

ERL — Emergent Retrieval Landmark 2026 – Present
Studying graph routing, evidence selection, and reader sensitivity for multi-hop question answering on LoCoMo, MuSiQue, and 2WikiMultihopQA. Initial routing matched BM25 on the held-out LoCoMo test; fixed-pool selection gains depend on the dataset and reader. Research dossier

RIFT — Recoverability-Gated Implicit Facial Dynamics 2026 – Present
Investigating recoverable motion and residual uncertainty in IMTalker's implicit motion space for audio-driven talking heads. A completed 36-fit temporal pilot did not support the tested hypothesis; representation-fidelity and residual-dynamics studies are planned before generative modeling. Research dossier

Publications and Manuscripts

[1] Mahir Shahriar Tamim, Md. Samiul Alim, **Tanvir Ahmed Khan**, Maisha Rahman, and Md Musfique Anwar. "Ensemble and Temporal Feature-Based Framework for Rainfall Classification in Bangladesh." *PLOS ONE*, 21(3): e0342646, 2026. DOI.

[2] Mahir Shahriar Tamim, Sharjil Khan, Md. Samiul Alim, **Tanvir Ahmed Khan**, Shafin Rahman, and Nabeel Mohammed. "CAT-GS: Balanced Multimodal Learning via Calibrated Gating and Fusion Surgery." Manuscript under minor revision at *Neurocomputing*, 2026. arXiv.

[3] Md. Samiul Alim, Mahir Shahriar Tamim, **Tanvir Ahmed Khan**, Sharjil Khan, Rafia Ferdous Duti, Shahriyar Zaman Ridoy, and Mohammad Ali Moni. "Can LLMs Follow the Pulse of a Crisis? Evaluating Crisis Sentiment in Bangladesh's July Uprising." *AACL-IJCNLP 2026*, Main Conference, accepted, 2026. OpenReview; arXiv.

[4] Md. Samiul Alim, Mahir Shahriar Tamim, Maisha Rahman, **Tanvir Ahmed Khan**, and Md Mushfique Anwar. "When a Nation Speaks: Machine Learning and NLP in People's Sentiment Analysis During Bangladesh's 2024 Mass Uprising." In *2025 28th International Conference on Computer and Information Technology (ICCIT)*, pp. 413–418, IEEE, 2025. DOI.

Research Experience

Machine Intelligence Lab (MILab) | North South University Mar 2025 – Present

- **Research Assistant**. Advisors: Dr. Nabeel Mohammed and Dr. Shafin Rahman.
- Contribute to research in multimodal learning, low-resource NLP, and speech-driven visual systems, including the publications and manuscript listed above.
- Developed distributed multi-GPU inference and chunked execution for collaborative talking-avatar systems; investigated Audio2Face–Three.js integration and ARKit blend-shape recovery.
- Contributed to synchronized streaming, renderer distillation, TensorRT validation, and Jetson profiling using external speech and rendering models; portions of the work are closed source.
- Profiling identified upstream speech generation as the bottleneck after renderer acceleration; end-to-end real-time operation was not established. Archived a separate 3D Gaussian method hypothesis after its registered residual-association gate failed.

Selected Research and Engineering Projects

PNEUMA — Video Memory for Long-Form Question Answering Discontinued thesis work

Discontinued undergraduate thesis research from a four-person team, integrating frozen perception models with SQL, vector, and temporal-graph memory. An evaluation audit exposed unresolved grounding and correctness limitations. Research dossier

Saliency Detection via Knowledge Distillation Aug 2024 – Dec 2024

Knowledge-distillation prototype using a Peekaboo/DINO teacher and UKAN student with mask-supervised and teacher–student objectives. GitHub

DynApex: Dynamic CPU/GPU Parallelization Aug 2024 – Dec 2024

C++ prototype for dependency and cycle analysis, pragma injection, and OpenMP-based CPU and target execution. GitHub

Industry Experience

Tuctuc | Spain (Remote) Jan 2022 – Jun 2026

- **Junior Software Engineer**. Maintained and extended Odoo-based ERP systems and developed cross-platform mobile applications using React Native.

Technical Skills

Machine Learning: PyTorch, Hugging Face Transformers, Knowledge Distillation, Multimodal Fusion

Vision & Avatar Systems: 3D Gaussian Splatting, FLAME/3DMM, NVIDIA Audio2Face, ARKit Blend Shapes

Retrieval & Memory: Retrieval-Augmented Generation, FAISS, SQL, Temporal Graphs, Sentiment Analysis

Systems & Deployment: Multi-GPU Inference, TensorRT, Docker, Linux, NVIDIA Jetson AGX Orin, FastAPI, Git

Programming: Python, C++, JavaScript, React Native

Languages: Bangla (Native), English (C1)

Awards & Honors

Academic Scholarship, North South University 2023

Participant, ICPC Asia Dhaka Preliminary Contest 2021